통계열역학 (2014년도 1학기)

과제 #1 정답 (총 **110** 점)

1. **(총 30점, 각 5점)**

(a) $Q(N,V,T) = 3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}$ 이므로,
미시상태 "1" 에 존재할 확률은

$$\begin{aligned} p_1 &= \frac{e^{-E_a/(k_BT)}}{Q(N,V,T)} \\ &= \frac{e^{-E_a/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \end{aligned}$$

(b) 에너지준위 "1" 의 에너지는 E_a 이므로, 이 에너지 준위에 존재할 확률은

$$\begin{aligned} p_{E_a} &= \frac{3e^{-E_a/(k_BT)}}{Q(N,V,T)} \\ &= \frac{3e^{-E_a/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \end{aligned}$$

(c) 미시상태 "4" 의 에너지는 E_c 이므로 이 미시상태에 존재할 확률은 다음과 같다. 미시상태 "4" 에 존재할 확률은

$$\begin{aligned} p_4 &= \frac{e^{-E_c/(k_BT)}}{Q(N,V,T)} \\ &= \frac{e^{-E_c/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \end{aligned}$$

(d) 각 미시상태에 존재할 확률을 모두 더하면

$$\begin{aligned} & p_1 + p_2 + p_3 + p_4 + p_5 + p_6 + p_7 + p_8 + p_9 + p_{10} \\ &= \frac{e^{-E_a/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} + \frac{e^{-E_b/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \\ &\quad + \frac{e^{-E_b/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} + \frac{e^{-E_c/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \\ &\quad + \frac{e^{-E_b/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} + \frac{e^{-E_a/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \\ &\quad + \frac{e^{-E_b/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} + \frac{e^{-E_c/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \\ &\quad + \frac{e^{-E_c/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} + \frac{e^{-E_a/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \\ &= \frac{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} = 1 \end{aligned}$$

(e) 각 에너지준위에 존재할 확률을 모두 더하면

$$\begin{aligned} & p_{E_a} + p_{E_b} + p_{E_c} \\ &= \frac{3e^{-E_a/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} + \frac{4e^{-E_b/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \\ &\quad + \frac{3e^{-E_c/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \\ &= \frac{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} = 1 \end{aligned}$$

(f) 평균에너지는 다음과 같으므로

$$\langle E\rangle = \frac{3E_a e^{-E_a/(k_BT)} + 4E_b e^{-E_b/(k_BT)} + 3E_c e^{-E_c/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}}$$

일정부피 몰 열용량은

$$\begin{aligned}
\bar{C}_V(T) &= \frac{\partial\langle E\rangle}{\partial T} \\
&= \frac{3E_a e^{-E_a/(k_BT)} + 4E_b e^{-E_b/(k_BT)} + 3E_c e^{-E_c/(k_BT)}}{3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)}} \\
&= \frac{1}{k_BT^2}\left[\frac{(3E_a^2 e^{-E_a/(k_BT)} + 4E_b^2 e^{-E_b/(k_BT)} + 3E_c^2 e^{-E_c/(k_BT)})(3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)})}{(3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)})^2}\right. \\
&\quad \left. - \frac{(3E_a e^{-E_a/(k_BT)} + 4E_b e^{-E_b/(k_BT)} + 3E_c e^{-E_c/(k_BT)})^2}{(3e^{-E_a/(k_BT)} + 4e^{-E_b/(k_BT)} + 3e^{-E_c/(k_BT)})^2}\right]
\end{aligned}$$

2. **(총 25점, 각 5점)**

(a) 전기장 B_z 하에서 ^{1}H 한 개로 구성되어 있는 계의 핵스핀에 의한 에너지는 다음과 같이 주어지므로

$$E_{\pm\frac{1}{2}} = \mp\frac{1}{2}\hbar\gamma B_z$$

정준분배함수는 다음과 같이 쓸 수 있다.

$$\begin{aligned}
Q(T, B_z) &= e^{\beta\hbar\gamma B_z/2} + e^{-\beta\hbar\gamma B_z/2} \\
&= e^{\hbar\gamma B_z/(2k_BT)} + e^{-\hbar\gamma B_z/(2k_BT)}
\end{aligned}$$

(b) 평균에너지,즉 내부에너지 $\langle E\rangle$ 는 다음과 같이 구할 수 있다.

$$\begin{aligned}
\langle E\rangle &= -\left(\frac{\partial \ln Q}{\partial\beta}\right)_{N,V}, \\
&= -\frac{\hbar\gamma B_z}{2}\left(\frac{e^{\beta\hbar\gamma B_z/2} - e^{-\beta\hbar\gamma B_z/2}}{e^{\beta\hbar\gamma B_z/2} + e^{-\beta\hbar\gamma B_z/2}}\right), \\
&= -\frac{\hbar\gamma B_z}{2}\left(\frac{e^{\hbar\gamma B_z/2k_BT} - e^{-\hbar\gamma B_z/2k_BT}}{e^{\hbar\gamma B_z/2k_BT} + e^{-\hbar\gamma B_z/2k_BT}}\right)
\end{aligned}$$

(c) 일정부피 열용량은 다음과 같이 구할 수 있다.

$$\begin{aligned}
C_V &= \left(\frac{\partial\langle E\rangle}{\partial T}\right)_V, \\
&= -\frac{\hbar\gamma B_z}{2}\left(\frac{(e^{\hbar\gamma B_z/2k_BT} + e^{-\hbar\gamma B_z/2k_BT})^2\left(-\frac{\hbar\gamma B_z}{2k_BT^2}\right) - (e^{\hbar\gamma B_z/2k_BT} - e^{-\hbar\gamma B_z/2k_BT})^2\left(-\frac{\hbar\gamma B_z}{2k_BT^2}\right)}{(e^{\hbar\gamma B_z/2k_BT} + e^{-\hbar\gamma B_z/2k_BT})^2}\right) \\
&= \frac{(\hbar\gamma B_z)^2}{k_BT^2}\frac{1}{(e^{\hbar\gamma B_z/2k_BT} + e^{-\hbar\gamma B_z/2k_BT})^2}. \qquad (1)
\end{aligned}$$

(d) $E_\alpha = -\frac{1}{2}\hbar\gamma B_z$ 이고, $E_\beta = \frac{1}{2}\hbar\gamma B_z$ 이므로,

$$\begin{aligned} p_\alpha &= \frac{e^{\hbar\gamma B_z/(2k_BT)}}{Q(B_z,T)} \\ &= \frac{e^{\hbar\gamma B_z/(2k_BT)}}{e^{\hbar\gamma B_z/(2k_BT)} + e^{-\hbar\gamma B_z/(2k_BT)}} \end{aligned} \tag{2}$$

$$\begin{aligned} p_\beta &= \frac{e^{-\hbar\gamma B_z/(2k_BT)}}{Q(B_z,T)} \\ &= \frac{e^{-\hbar\gamma B_z/(2k_BT)}}{e^{\hbar\gamma B_z/(2k_BT)} + e^{-\hbar\gamma B_z/(2k_BT)}} \end{aligned} \tag{3}$$

한편,

$$\frac{\hbar\gamma B_z}{2k_BT} = \frac{(1.0546\times10^{-34}\ \text{J s})(26.75\times10^{7}\ \text{T}^{-1}\ \text{s}^{-1})(12\ \text{T})}{(2)(1.38066\times10^{-23}\ \text{J K}^{-1})(300\ \text{K})} = 4.09\times10^{-5} \tag{4}$$

그러므로, (2) 식은

$$p_\alpha = \frac{e^{4.09\times10^{-5}}}{e^{4.09\times10^{-5}} + e^{-4.09\times10^{-5}}} = 0.5$$

같은 방법으로 (3) 식은

$$p_\beta = \frac{e^{-4.09\times10^{-5}}}{e^{4.09\times10^{-5}} + e^{-4.09\times10^{-5}}} = 0.5$$

(e) $T = 10.0$ K 인 경우,

$$\frac{\hbar\gamma B_z}{2k_BT} = \frac{(1.0546\times10^{-34}\ \text{J s})(26.75\times10^{7}\ \text{T}^{-1}\ \text{s}^{-1})(12\ \text{T})}{(2)(1.38066\times10^{-23}\ \text{J K}^{-1})(10.0\ \text{K})} = 1.23\times10^{-3} \tag{5}$$

그러므로, (2) 식은

$$p_\alpha = \frac{e^{1.23\times10^{-3}}}{e^{1.23\times10^{-3}} + e^{-1.23\times10^{-3}}} = 0.5006$$

같은 방법으로 (3) 식은

$$p_\beta = \frac{e^{-1.23\times10^{-3}}}{e^{1.23\times10^{-3}} + e^{-1.23\times10^{-3}}} = 0.4994$$

3. **(총 25점, 각 5점)**

(a)

$$Q(N,V,T) = 1 + e^{-E_0/(k_BT)} + \gamma e^{-2E_0/(k_BT)} \tag{6}$$

(b)

$$\begin{aligned} \langle E\rangle &= k_BT^2\left(\frac{\partial \ln Q}{\partial T}\right)_{N,V} \\ &= k_BT^2\frac{\partial}{\partial T}\left[\ln\left(1 + e^{-E_0/(k_BT)} + \gamma e^{-2E_0/(k_BT)}\right)\right] \\ &= \frac{E_0e^{-E_0/(k_BT)} + 2E_0e^{-2E_0/(k_BT)}}{1 + e^{-E_0/(k_BT)} + \gamma e^{-2E_0/(k_BT)}} \end{aligned}$$

(c)

$$p_1 = \frac{1}{1 + e^{-E_0/(k_B T)} + \gamma e^{-2E_0/(k_B T)}} \tag{7}$$

$$p_2 = \frac{e^{-E_0/(k_B T)}}{1 + e^{-E_0/(k_B T)} + \gamma e^{-2E_0/(k_B T)}} \tag{8}$$

$$p_3 = \frac{\gamma e^{-2E_0/(k_B T)}}{1 + e^{-E_0/(k_B T)} + \gamma e^{-2E_0/(k_B T)}} \tag{9}$$

$E_0/(k_B T) = 1$, $\gamma = 1$ 0이므로, (12) 식, (13) 식, (14) 식으로부터

$$p_1 = \frac{1}{1 + e^{-1} + e^{-2}} = 0.665$$
$$p_2 = \frac{e^{-1}}{1 + e^{-1} + e^{-2}} = 0.245$$
$$p_3 = \frac{e^{-2}}{1 + e^{-1} + e^{-2}} = 0.090$$

(d) 온도가 T_0일 때, $p_1 = p_3$ 이므로 (12) 식과 (14) 식으로부터

$$1 = \gamma e^{-2E_0/(k_B T_0)} \Rightarrow T_0 = \frac{2E_0}{k_B \ln \gamma} = \frac{(2)(2 \text{ kcal mol}^{-1})}{(1.38066 \times 10^{-23} \text{ J K}^{-1})(\ln 1000)} \tag{10}$$
$$= \frac{(2)(2 \times 4.184 \times 10^3 \text{ J})}{(6.022 \times 10^{23})(1.38066 \times 10^{-23} \text{ J K}^{-1})(\ln 1000)}$$
$$= 291.4 \text{ K}$$

(e) (10) 식으로부터

$$\frac{E_0}{k_B T_0} = \frac{\ln \gamma}{2} = \frac{\ln 1000}{2} = 3.454$$

0이므로

$$p_3 = p_1 = \frac{1}{1 + e^{-E_0/(k_B T)} + \gamma e^{-2E_0/(k_B T)}}$$
$$= \frac{1}{1 + e^{-3.454} + 1000 \times e^{-(2)(3.454)}} = 0.492$$
$$p_2 = \frac{e^{-E_0/(k_B T)}}{1 + e^{-E_0/(k_B T)} + \gamma e^{-2E_0/(k_B T)}}$$
$$= \frac{e^{-3.454}}{1 + e^{-3.454} + 1000 \times e^{-(2)(3.454)}} = 0.0156$$

4. **(총 30점)**

(a) **(5점)**

$$Q(2, V, T) = \sum_\sigma \Omega(E_\sigma) e^{-\beta E_\sigma}$$
$$= \underbrace{(V-1)e^{\beta\epsilon}}_{\text{이량체에 대한 분자분배함수}} + \underbrace{\left(\frac{V}{2} - 1\right)(V-1)}_{\text{단량체에 대한 분자분배함수}} \tag{11}$$

(b) **(5점)**

$$\begin{aligned} p_{\text{dimer}} &= \frac{(V-1)e^{\beta\epsilon}}{Q(2,V,T)} \\ &= \frac{e^{\beta\epsilon}}{e^{\beta\epsilon} + \left(\frac{V}{2}-1\right)} \end{aligned} \tag{12}$$

$$\begin{aligned} p_{\text{monomer}} &= \frac{\left(\frac{V}{2}-1\right)(V-1)}{Q(2,V,T)} \\ &= \frac{\left(\frac{V}{2}-1\right)}{e^{\beta\epsilon} + \left(\frac{V}{2}-1\right)} \end{aligned} \tag{13}$$

(c) **(5점)** $p_{\text{dimer}} = p_{\text{monomer}}$ 이므로 (b)의 (12) 식과 (13) 식으로로부터

$$e^{\beta\epsilon} = \frac{V}{2} - 1 \Rightarrow \beta\epsilon = \ln\left(\frac{V}{2}-1\right) \tag{14}$$

$\beta = 1/k_B T_0$ 이므로 (14) 식으로부터

$$T_0 = \frac{\epsilon}{k_B \ln(V/2-1)} \tag{15}$$

(d) **(5점)**

$$A_{\text{tot}} = -k_B T \ln Q(2,V,T) = -k_B T \ln\left[(V-1)e^{\beta\epsilon} + \left(\frac{V}{2}-1\right)(V-1)\right] \tag{16}$$

(e) **(10점)** 수업시간에 배운 결과에 의하면

$$A_{\text{monomer}} = -k_B T \ln\left[\left(\frac{V}{2}-1\right)(V-1)\right]$$

$$A_{\text{dimer}} = -\epsilon - k_B T \ln(V-1)$$

20 kcal/mol $= 1.389 \times 10^{-19}$ 이므로

